

Introduction

Background:

Microplastics are plastic particles **less than 5 mm** in size that have become widespread **environmental pollutants** in freshwater ecosystems (Koelmans et al., 2022). These particles can be **ingested** by aquatic organisms during feeding, **accumulating in digestive systems** and potentially causing **physiological stress** (Kong et al., 2020). Research shows microplastics can **impair growth, reproduction, and survival** in various aquatic invertebrates (Guo et al., 2022).

Lumbricus variegatus (blackworms) are freshwater annelids known for their **regenerative capabilities**, making them **ideal** model organisms for studying toxicological impacts (Silva et al., 2021). When injured, blackworms can **regenerate posterior segments** through a complex process requiring significant **metabolic energy**.

The Problem:

- Microplastic pollution in freshwater systems is **increasing worldwide**.
- Effects on regenerative organisms are **poorly understood**.
- No current methods exist** to mitigate **internal** microplastic accumulation.

Novel Approach:

This research introduces **sodium-alginate hydrogels** (biocompatible materials with porous structures) as a potential **internal** treatment to **bind** ingested microplastics and **facilitate their excretion**.

Objectives

Research Question: How do varying **microplastic concentrations** affect posterior **regeneration** in *Lumbricus variegatus*, and can sodium-alginate hydrogels **mitigate** these effects?

Independent Variable:

- Microplastic **concentration** (10, 100, 1000 particles/mL)
- Presence or absence** of sodium alginate hydrogel beads (5 beads per well plate)

Dependent Variable:

Posterior regeneration time, measured in number of days required for amputated worms to **generate at least 5 posterior segments** post-amputation.

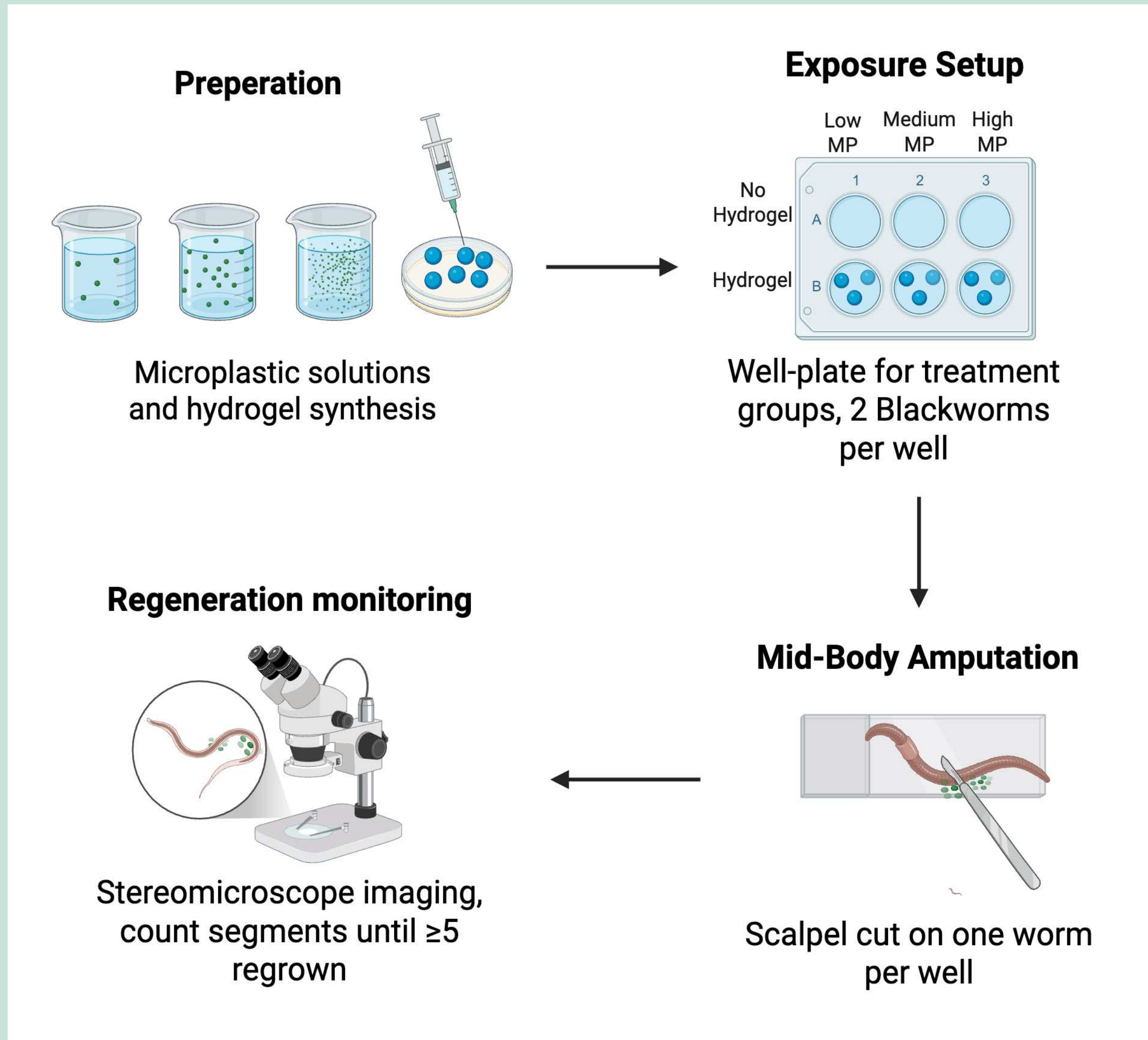
Hypothesis:

- Increasing microplastic concentration will:
- Slow** posterior regeneration
- Reduce** total segments regenerated
- These negative effects will **be reduced by hydrogel treatment**

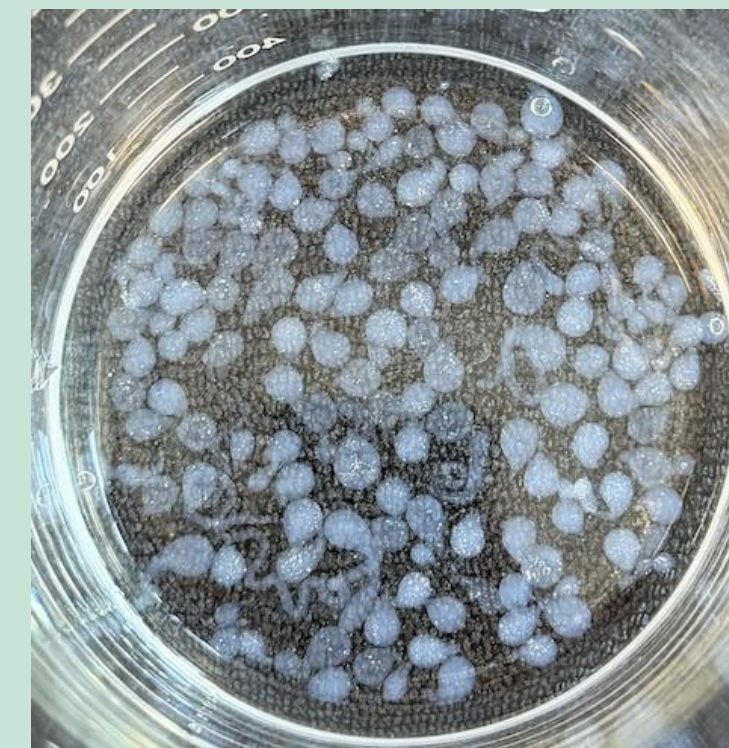
Justification: Microplastics cause **oxidative stress** and **physical gut obstruction**, diverting **metabolic energy** away from **tissue repair** toward detoxification (Zimmermann et al., 2020). Sodium alginate hydrogels possess negatively charged functional groups that can **bind microplastic particles** through electrostatic interactions (Han et al., 2025), potentially **reducing microplastic bioavailability** within the digestive system and allowing organisms to **allocate more energy toward regeneration**.

Evaluating the impact of microplastic ingestion and hydrogel treatment on the growth and regeneration in blackworms (*Lumbricus variegatus*)

Methodology



Experimental design visualization created by student researcher using BioRender.



Sodium alginate hydrogel beads synthesized for experiment. Image taken by student researcher.

- Hydrogel synthesis:** Sodium alginate + CaCl_2 crosslinking
- Microplastics:** Fluorescent polyethylene beads (50-100 μm)
- Treatment setup:** 2 trials $\times$ 5 worms per group
- Mid-body Amputation:** Anterior segment removal (Day 0)
- Measurements:** Days 6, 8, 17, 19 post-amputation, imaging processed through ImageJ
- Data collected:** Regeneration length (mm), segment count, survival



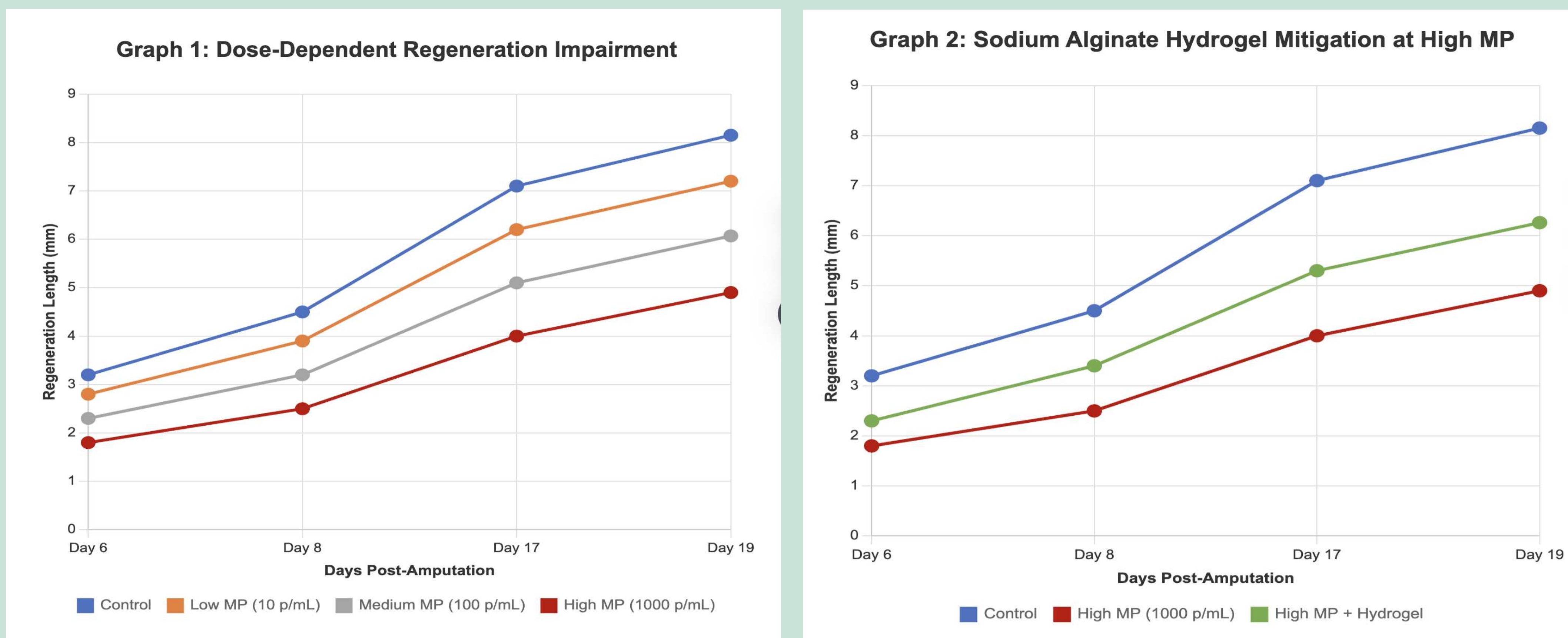
Blackworm day 1 post amputation



Blackworm day 8 post amputation

Blackworm specimens. Images taken by student researcher using stereomicroscope.

Results



Graphs generated by student researcher using Microsoft Excel with data from experimental measurements.

Findings:

- Dose-Response Relationship:** regeneration impairment increases proportionally with microplastic concentration.
- The sodium alginate hydrogel successfully improved regeneration at the highest microplastic level by 27.8%.
- Structural Regeneration Impaired:** High microplastic exposure reduced segment formation from 9.2 to 5.0 segments.
- Survival Impact Reduced:** High microplastic concentration caused 20% mortality (2 out of 10 worms), while hydrogel treatment reduced mortality to 10% (1 out of 10 worms)

Conclusion

Statistical Analysis:

One-way ANOVA revealed microplastic concentration significantly affected regeneration ($F(3,34) = 436.46$, $p < 0.05$), with all concentrations causing **significant impairment** compared to controls. Independent T-tests showed that alginate hydrogel significantly **improved regeneration** at all microplastic concentrations tested ($p < 0.05$), with strongest mitigation at high concentration (**27.8% improvement**). High microplastic exposure significantly **reduced segment formation** by 46%, indicating that **microplastics disrupt both regeneration speed and structural quality**.

Key Conclusions:

- Microplastics impair regeneration in a **concentration-dependent manner**. Even low concentrations (10 p/mL) caused significant harm.
- Sodium alginate hydrogel **significantly rescues regeneration** at all concentrations, with effectiveness increasing at higher microplastic levels.
- Structural Impairment:** Microplastics **disrupt both regeneration length and segment formation**, indicating **comprehensive damage** to regenerative processes.
- Environmentally relevant microplastic concentrations significantly threaten aquatic invertebrate populations by **impairing regenerative capabilities** critical for survival.

Hypothesis was shown to be **supported** as increased microplastic concentrations **slowed regeneration** and **reduced total segments regenerated**.

Statistical analysis confirms both **dose-dependent microplastic impairment** and **concentration-dependent hydrogel protection**.

References

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